

6502 Machine Language Hex Codes

Opcode	Definition	Immediate	Accumulator	Zero Page	Zero Page,X	Zero Page,Y	Absolute	Absolute,X	Absolute,Y	Indirect Abs.	(Zero Page,X)	(Zero Page),Y	Implied	Relative	Affects Flags
ADC	Add with Carry	69		65	75		6D	7D	79		61	71			N,V,Z,C
AND	Bitwise AND with Accumulator	29		25	35		2D	3D	39		21	31			N,Z
ASL	Arithmetic Shift Left		0A	06	16		0E	1E							N,Z,C
BCC	Branch on Carry Clear													90	
BCS	Branch on Carry Set													B0	
BEQ	Branch on Equal													F0	
BIT	Test Bits			24			2C								N,V,Z
BMI	Branch on Minus													30	
BNE	Branch on Not Equal													D0	
BPL	Branch on Plus													10	
BRK	Break												00		B
BVC	Branch on Overflow Clear													50	
BVS	Branch on Overflow Set													70	
CLC	Clear Carry												18		C
CLD	Clear Decimal												D8		D
CLI	Clear Interrupt												58		I
CLV	Clear Overflow												B8		V
CMP	Compare Accumulator	C9		C5	D5		CD	DD	D9		C1	D1			N,Z,C
CPX	Compare X Register	E0		E4			EC								N,Z,C
CPY	Compare Y Register	C0		C4			CC								N,Z,C
DEC	Decrement Memory			C6	D6		CE	DE							N,Z
DEX	Decrement X												CA		N,Z
DEY	Decrement Y												88		N,Z
EOR	Bitwise Exclusive OR	49		45	55		4D	5D	59		41	51			N,Z
INC	Increment Memory			E6	F6		EE	FE							N,Z
INX	Increment X												E8		N,Z
INY	Increment Y												C8		N,Z
JMP	Jump						4C			6C					
JSR	Jump to Subroutine						20								
LDA	Load Accumulator	A9		A5	B5		AD	BD	B9		A1	B1			N,Z
LDX	Load X Register	A2		A6		B6	AE		BE						N,Z
LDY	Load Y Register	A0		A4	B4		AC	BC							N,Z
LSR	Logical Shift Right		4A	46	56		4E	5E							N,Z,C
NOP	No Operation												EA		
ORA	Bitwise OR with Accumulator	09		05	15		0D	1D	19		01	11			N,Z
PHA	Push Accumulator to Stack												48		
PHP	Push Processor Status to Stack												08		
PLA	Pull Accumulator off Stack												68		N,Z
PLP	Pull Processor Status off Stack												28		ALL
ROL	Rotate Left		2A	26	36		2E	3E							N,Z,C
ROR	Rotate Right		6A	66	76		6E	7E							N,Z,C
RTI	Return from Interrupt												40		ALL
RTS	Return from Subroutine												60		
SBC	Subtract with Carry	E9		E5	F5		ED	FD	F9		E1	F1			N,V,Z,C
SEC	Set Carry												38		C
SED	Set Decimal												F8		D
SEI	Set Interrupt												78		I
STA	Store Accumulator			85	95		8D	9D	99		81	91			
STX	Store X Register			86		96	8E								
STY	Store Y Register			84	94		8C								
TAX	Transfer A to X												AA		N,Z
TAY	Transfer A to Y												A8		N,Z
TSX	Transfer Stack Pointer to X												BA		
TXA	Transfer X to A												8A		N,Z
TXS	Transfer X to Stack Pointer												9A		
TYA	Transfer Y to A												98		N,Z

The result of the opcode directly modifies....

System Memory (Gray 2)

8-bit Registers (Accumulator/X/Y) (Sky blue 10)

16-bit Registers (AB, CD, EF, etc.) (Magenta 10)

CPU Status Register (Orange 1)

Stack/Stack Pointer (Red 2)

Program Counter (Green 2)

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This table originally authored by Lawrence Woodman (https://github.com/lawrencewoodman/machine_language_aids)Opcode color coding and formatting changes by Brian Manning (https://github.com/spicyjack/8-16_bit_cpu_info)License: Creative Commons Attribution 4.0 International License (<https://creativecommons.org/licenses/by/4.0/>)

Numeric Order

Hex	Instruction	ASCII	Dec	Oct	Nyb H	Nyb L	Hex	Instruction	ASCII	Dec	Oct	Nyb H	Nyb L
00	BRK	NUL	0	000	0000	0000	00	BRK	NUL	0	000	0000	0000
01		SOH	1	001	0000	0001	01		SOH	1	001	0000	0001
02		STX	2	002	0000	0010	02			2	002	0000	0010
03		ETX	3	003	0000	0011	03			3	003	0000	0011
04		EOT	4	004	0000	0100	04			4	004	0000	0100
05		ENQ	5	005	0000	0101	05			5	005	0000	0101
06		ACK	6	006	0000	0110	06			6	006	0000	0110
07		BEL	7	007	0000	0111	07			7	007	0000	0111
08		BS	8	010	0000	1000	08			8	010	0000	1000
09		HT	9	011	0000	1001	09			9	011	0000	1001
0A		LF	10	012	0000	1010	0A			10	012	0000	1010
0B		VT	11	013	0000	1011	0B			11	013	0000	1011
0C		FF	12	014	0000	1100	0C			12	014	0000	1100
0D		CR	13	015	0000	1101	0D			13	015	0000	1101
0E		SO	14	016	0000	1110	0E			14	016	0000	1110
0F		SI	15	017	0000	1111	0F			15	017	0000	1111
10		DLE	16	020	0001	0000	10			16	020	0001	0000
11		DC1	17	021	0001	0000							
12		DC2	18	022	0001	0001							
13		DC3	19	023	0001	0010							
14		DC4	20	024	0001	0011							
15		NAK	21	025	0001	0100							
16		SYN	22	026	0001	0101							
17		ETB	23	027	0001	0110							
18		CAN	24	030	0001	0111							
19		EM	25	031	0001	1000							
1A		SUB	26	032	0001	1001							
1B		ESC	27	033	0001	1010							
1C		FS	28	034	0001	1011							
1D		GS	29	035	0001	1100							
1E		RS	30	036	0001	1101							
1F		US	31	037	0001	1110							
20		DEL	32	040	0001	1111							
						0000							

		Hex Matrix															
First Hex Digit		Second Hex Digit															
	0x0	BRK															
	0x1																
	0x2																
	0x3																
	0x4																
	0x5																
	0x6																
	0x7																
	0x8																
	0x9																
	0xA																
	0xB																
	0xC																
	0xD																
	0xE																
	0xF																